Project Synopsis:- Criminal Vehicle Detection & Number Plate Recognition System Using Deep Learning.

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Project Title :-Criminal Vehicle Detection and Automatic Number Plate Recognition Using YOLOv8 and OCR.

Introduction:

This project presents an AI-powered surveillance system designed to detect criminal or suspicious vehicles by automatically identifying license plate numbers from images or video streams. The system integrates vehicle detection, number plate extraction, and character recognition into a single intelligent pipeline.
Such systems are crucial for modern security infrastructures, including police surveillance, border control, restricted-area monitoring, and smart city crime prevention.

Objective:

- To develop an AI model capable of detecting vehicles accurately.
- To identify and extract number plates from detected vehicles.
- To read and recognize plate numbers using OCR.
- To match detected plate numbers against a database of criminal or flagged vehicles.
- To generate real-time alerts for suspicious activity.

System Architecture:

- Input (Image/Video/Live CCTV Feed)
- Vehicle Detection Module
- Number Plate Detection Module
- Plate Cropping & Pre-processing

- OCR Character Recognition
- Database Matching (Criminal Vehicle List)
- Alert Generation
- Output (Annotated Frames + Matched Details)

Methodology:

1. Data Collection & Annotation

A diverse dataset of vehicles and number plates was prepared and annotated to train the detection model effectively.

2. Model Training

- YOLOv8 trained for plate detection
- Augmentation for improving generalization
- Hyperparameter tuning for accuracy and speed

3. OCR Pipeline

The plate region undergoes preprocessing before OCR takes place:

- Noise reduction
- Grayscale conversion
- Thresholding
- Text extraction

4. Real-Time Integration

All modules were integrated into a single workflow capable of real-time inference.

5. Testing & Validation

The system was tested on challenging video sequences including fast-moving vehicles, lowlight footage, and complex traffic.

Tools and Technologies Used:

- Python
- YOLOv8 (Deep Learning)
- OCR Engine (EasyOCR / Tesseract)

- OpenCV
- NumPy, Pandas
- SQLite / JSON for Criminal Vehicle Database (if needed)
- Tracking Algorithm (SORT / Kalman Filter)

Advantages:

- Automated detection and alerting
- Real-time performance
- High accuracy due to deep learning
- Scalable for large surveillance setups
- Can be integrated with existing CCTV systems

Conclusion:

The Criminal Vehicle Detection and Automatic Number Plate Recognition system provides an intelligent, automated, and real-time solution for enhancing public safety and surveillance. By combining deep learning–based detection and OCR-based recognition, the system offers a reliable method for identifying suspicious vehicles and supporting law enforcement in crime prevention and investigation.